

1. Alice rolls a 6-sided dice (numbered 1 to 6), and Bob rolls a 20-sided dice (numbered 1 to 20).

Let X denote the minimum of the two values (that appear on their two dice). What is the probability mass function of X ?

2. Alice flips a fair coin until the first time that she gets heads, and then she stops. Let X denote the number of flips that Alice makes (including the final coin flip, which is heads).

2a. What is $P(X \leq 3)$?

2b. What is $P(X \leq 10)$?

2c. Fix an integer $x \geq 1$. What is $P(X \leq x)$?

3. Suppose that $P(Y = y) = (1/2)^y$ for integers $y \geq 1$.

3a. Find the probability that Y is even, i.e., that Y is a multiple of 2.

3b. Find the probability that Y is a multiple of 3.

3c. Find the probability that Y is a multiple of 4.

3d. Fix an integer $r \geq 1$. Find the probability that Y is a multiple of r .

4. Consider a discrete random variable X with probability mass function $f_X(x) = (3/5)^{x-1}(2/5)$ for $x \geq 1$.

Fix an integer $r \geq 0$. Find the value of $P(X > r)$.

5. Consider a discrete random variable Y with probability mass function $f_Y(y) = (2/7)^{y-1}(5/7)$ for $y \geq 1$.

Let a and b be fixed positive integers with $a < b$.

Calculate the value of $P(a \leq Y \leq b)$.